



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 6
Document Code: WI-01-MIC-05	STAPHYLOCOCCUS AUREUS IN FISH AND FISHERY PRODUCTS		

1. OBJECTIVE

To determine the number of *Staphylococcus aureus* present in fish and fishery products using Direct Plate Count Method.

2. SCOPE

This instruction is applicable in the determination of total number of *Staphylococcus aureus* in fish and fishery products.

3. RESPONSIBILITY

Laboratory Analyst.

4. APPARATUS/EQUIPMENT/CONSUMABLES

- 4.1 Alcohol lamp / Fireboy
- 4.2 Disinfectant (70% solution isopropyl alcohol)
- 4.3 Autoclave and Autoclave tape
- 4.4 Forceps – stainless steel
- 4.5 Scissors – stainless steel
- 4.6 Knives / Can opener
- 4.7 Chopping board
- 4.8 Top loading balance
- 4.9 Sterile stomacher bags
- 4.10 Stomacher
- 4.11 Aluminum foil
- 4.12 Tissue paper / paper towel
- 4.13 Graduated cylinders (100 and 500 ml, Pyrex)
- 4.14 Dilution bottles with screw caps (160 ml, Pyrex)
- 4.15 Media bottles (500 ml or 1000 ml, Pyrex)

Prepared by: CHARISSE ZIANNE N. LIBRES Laboratory Analyst- Microbiology	Reviewed by: MA. MAURICE A. BANQUILING Quality Assurance Manager	Approved by: EUGENE GAY B. JAMORA Laboratory Manager
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- 4.16 Sterile pipettes (10 ml)
- 4.17 Sterile pipette tips (10 – 200 µl)
- 4.18 Sterile Petri dishes (100 x 115 mm)
- 4.19 Incubator maintained at 35°C
- 4.20 Spreader
- 4.21 Sterile inoculating needles
- 4.22 Glass-marking pen
- 4.23 Biosafety Cabinet
- 4.24 **Microscope**
- 4.25 **Colony Counter**

5. MEDIA AND REAGENTS:

5.1 All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

5.2 Baird – Parker Basal Medium

Suspend 60 g in 950 ml of demineralized water by heating in a boiling water bath or in a current of steam. Autoclave 15 min at 121°C. Final pH, 7.0 ± 0.2. If desired for immediate use, maintain melted medium at 48-50°C before adding enrichment. Otherwise, store solidified medium at 4 ± 1°C up to 1 month. Melt medium before use.

Complete medium

Aseptically add 5 ml prewarmed (45-50°C) BactoEggYolk Tellurite enrichment to 95 ml melted base. Mix well (avoiding bubbles) and pour 15-18 ml portions into sterile 15 x 100 mm petri dishes. The medium must be densely opaque. Dry plates before use. Store prepared plates at 20-25°C for up to 5 days.

5.3 Butterfield's Phosphate-Buffered Dilution Water

Suspend 34 g of KH₂PO₄ to 500 milliliter of distilled water. Adjust pH to 7.2 with 1 N NaOH. Bring volume to 1 liter with distilled water. Sterilize 15 min at 121°C. Store in a refrigerator at 4°C.

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Dilution Blanks

Take 1.25 ml of above stock solution and bring volume to 1 liter with distilled water.
Dispense into bottles to 90 or 99 ± 1 ml. Sterilize 15 min at 121°C .

5.4 Brain Heart Infusion (BHI) Broth

Suspend 37 g in a liter of demineralized water. Dispense into smaller vessels.
Autoclave for 15 minutes at 121°C . Final pH, 7.4 ± 0.2 .

5.5 Trypticase (tryptic) Soy Agar (TSA)

Suspend 40 g in a liter of demineralized water by heating in a boiling water bath or in a current of steam. Autoclave for 15 minutes at 121°C . Final pH, 7.3 ± 0.2 .

5.6 Coagulase plasma (rabbit) with EDTA

Reconstitute one vial of coagulase plasma with EDTA by adding 3 ml sterile distilled water. Shake well.

6. PROCEDURE:

6.1 Isolation and Enumeration of *Staphylococcus aureus*

- 6.1.1 Weigh 25 g sample into a sterile stomacher bag. Add 225 ml sterile Butterfield's phosphate-buffered dilution water and homogenize for 1 minute.
- 6.1.2 Prepare decimal dilutions of food homogenate such as 10^{-1} , 10^{-2} and 10^{-3} or as required by transferring 10 ml suspension into 90 ml diluent. Use a separate sterile pipette for making each transfer. Shake all dilutions 25 times in 30 cm arc within 7 seconds immediately prior to transfer to ensure uniform distribution of microorganisms.
- 6.1.3 For each dilution to be plated, aseptically transfer 1 ml sample suspension to 3 plates of Baird-Parker agar, distributing 1 ml of inoculum equitably to 3 plates (e.g., 0.4 ml, 0.3 ml, and 0.3 ml).

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- 6.1.4 Using a sterile bent glass streaking rod spread the inoculum evenly over the agar surface.
- 6.1.5 Let the plates stand until the inoculum has been absorbed completely by the agar. If inoculum is not readily absorbed, place plates upright in an incubator for about 1 h.
- 6.1.6 Invert the plates and incubate at 35°C for 45 - 48 h.
- 6.1.7 Select plates with 20 – 200 colonies, unless only plates at lower dilutions (>200 colonies) have colonies with typical appearance of *S. aureus*. Colonies of *S. aureus* are circular, smooth, convex, moist, 2-3 mm in diameter on uncrowded plates, gray to jet black, frequently with light-colored (off-white) margin, surrounded by opaque zone and frequently with an outer clear zone; colonies have buttery to gummy consistency when touched with inoculating needle. Strains isolated from frozen or desiccated foods that have been stored for extended periods frequently develop less black coloration than typical colonies and may have rough appearance and dry appearance and dry texture.
- 6.1.8 Count and record colonies. If several type of colonies are observed which appear to be *S.aureus* on selected plates, count number of colonies of each type and record counts separately. When plates of the lowest dilution contain <20 colonies, these may be used. If plates containing >200 colonies have colonies with the typical appearance of *S. aureus* and typical colonies do not appear at higher dilutions, use these plates for the enumeration of *S.aureus* but do not count non-typical colonies. Select >1 colony of each type counted and test for coagulase production.
- 6.1.9 Add number of colonies on triplicate plates represented by colonies giving positive coagulase test and multiply by the sample dilution factor. Report this number as number of *S. aureus/g* of food tested.

6.2 Coagulase Test

- 6.2.1 Transfer suspected *S. aureus* colonies into small tubes containing 0.2 - 0.3 ml BHI broth and emulsify thoroughly. Inoculate agar slant of suitable maintenance medium, e.g. TSA with loopful of BHI suspension.

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- 6.2.2 Incubate BHI culture suspension and slants 18-24 h at 35°C. Retain slant cultures at room temperature for ancillary or repeat tests in case coagulase test results are questionable.
- 6.2.3 Add 0.5 ml reconstituted coagulase plasma with EDTA to the BHI culture and mix thoroughly.
- 6.2.4 Incubate at 35°C and examine periodically over 6 h periods for clot formation. Only firm and complete clot that stays in place when tube is tilted or inverted is considered positive for *S. aureus*. Partial clotting, formerly 2+ and 3+ coagulase reactions must be tested further.
- 6.2.5 Test known positive and negative cultures simultaneously with suspect cultures of unknown coagulase activity.
- 6.2.6 Stain all suspect cultures with Gram reagent and observe microscopically.

6.3 Ancillary Test

❖ Catalase Test

Use growth from TSA slant for catalase test on glass slide or spot plate, and illuminate properly to observe production of gas bubbles. Pour 1 ml 3% hydrogen peroxide over growth on slant culture. Gas bubbles indicate positive test. Alternatively, emulsify colony in 1 drop 3% hydrogen peroxide on glass slide. Immediate bubbling is a positive catalase test.

Note: *Staphylococcus aureus* gives typical characteristics such as positive (+) coagulase production and positive (+) catalase activity.

7. REFERENCES

Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 3. Authors: Larry Maturin and James T. Peeler

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8. FORMS

LF W01-01, Reagent and Media Preparation Logbook

LF W01-MIC-03, Analyst Worksheet for Bacteriological Examination of Fish and Fishery Products

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